

Session-2 (TASKS)

1. Install GCC on your computer and verify the installation by running `gcc --version` in your terminal or command prompt.

Ans.

GCC (GNU Compiler Collection) is one of the most popular compilers used for programming languages such as C, C++, Objective-C, and others. Before writing and running a C program, it is necessary to install GCC on the computer.

Steps to Install GCC: -

For Windows: -

1. Download and install MinGW-w64 or another GCC package.
2. During installation, select the GCC compiler components.
3. Add the bin folder (for example, C:\MinGW\bin) to the system's Environment Variables (PATH).
4. Open Command Prompt.

For Linux: -

- Open the terminal and use the following command: -
`sudo apt install gcc`

For macOS: -

- Install the Xcode Command Line Tools by running:
`xcode-select --install`

Verifying the Installation: -

- After installation, open the Terminal or Command Prompt and type:
`gcc --version`
- If GCC has been installed correctly, information similar to the following will appear:
`gcc (GCC) 14.2.0`
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This is free software; see the source for copying conditions.
There is NO warranty.

2. Open VS Code or DevC++ and create a new file named `hello.c`. Write a C program that prints 'Hello, Instagram World!' to the console.

Ans.

Open VS Code or DevC++ and create a new file named: -

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hello.c

Write the following C program: -

```
#include<stdio.h>
int main()
{
    printf("Hello, Instagram World!");

    return 0;
}
```

Explanation of the Program: -

- #include<stdio.h> includes the Standard Input Output library.
- int main() is the starting point of program execution.
- printf() is used to display text on the screen.
- return 0; indicates that the program has executed successfully.

Expected Output: -

Hello, Instagram World!

3. Identify and label the header, main function, and return statement in your hello.c program using comments.

Ans.

The program with proper comments is shown below: -

```
// Header File
#include<stdio.h>
// Main Function
int main()
{
    printf("Hello, Instagram World!");

    // Return Statement
    return 0;
}
```

Description of Each Part: -

(A) Header File: -

```
#include<stdio.h>
```

The header file contains declarations for standard input and output functions. The printf() function is defined inside this header file.

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(B) Main Function: -

```
int main()
{
    ...
}
```

The main() function is the entry point of every C program. Execution always begins from this function.

(C) Return Statement: -

```
return 0;
```

The return statement ends the execution of the program and returns the value 0 to the operating system, indicating successful completion.

**4. Compile your hello.c file using the GCC command line (gcc hello.c -o hello) and run the output file to display the message in your terminal.

Hint: On Windows, the output file will be hello.exe; on Linux/Mac, it will be hello.**

Ans.

After saving the file, open the terminal or command prompt and navigate to the folder where hello.c is stored.

Step 1: Compile the Program: -

Use the following command: -

```
gcc hello.c -o hello
```

Explanation

- gcc → Calls the GCC compiler.
- hello.c → Source code file.
- -o hello → Creates an output file named hello.

Step 2: Run the Program: -

On Windows: -

```
hello.exe
```

or

```
.\hello.exe
```

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On Linux/macOS: -

```
./hello
```

Terminal Example: -

```
C:\Users\Student> gcc hello.c -o hello  
C:\Users\Student> hello.exe  
Hello, Instagram World!
```

Linux/macOS Example: -

```
$ gcc hello.c -o hello  
$ ./hello  
Hello, Instagram World!
```